

Carbon-Aware and Cost-Effective Geo-Distribution for Latency-Critical Microservices

Georgia Christofidi^{1,2}, Francisco Álvarez Terribas³, Ioannis Roumpos^{1,2}, Nicolas Kourtellis⁴, Jesus Alberto Omaña Iglesias³, Thaleia Dimitra Doudali¹

IMDEA Software Institute¹, Universidad Politécnica de Madrid², Telefonica Research³, Keysight Technologies⁴

A. Problem Space: Carbon-Aware Microservices (MS)

Problem: How can a MS company reduce its carbon emissions and infrastructure cost?

Existing Approaches
(Only for Global Scale): **Workload shifting**

Temporal: Wait until energy is greener.



(latency critical apps: not applicable)
Spatial: Move to a greener region. 🌱 🤔
(strict budgets: can be applied but carefully)

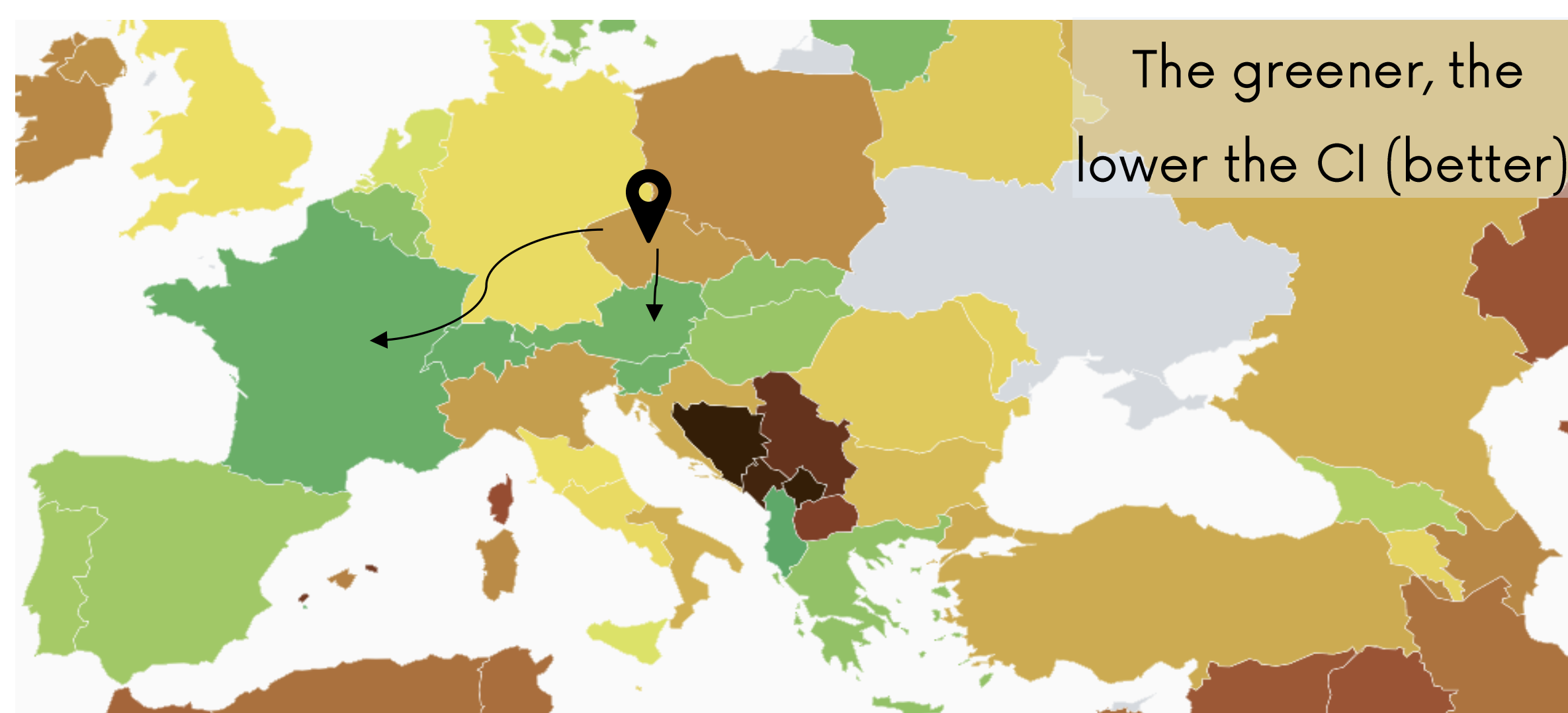


Fig. 1: Carbon intensity (CI) across different countries.

Use case: Small to medium-sized (SME) taxi reservation company in Czechia.



Harder Problem
Can a geo-constrained SME reduce both carbon and cost?

B. Insight: Move Early-Activated MS to Greener Regions

Goal: Map MS ↔ regions for

↓ Carbon ↓ Cost 🤔 Latency

Challenge: Which MS to move?
Where to move them?

⚠ Explosive Combinatorial Search Space

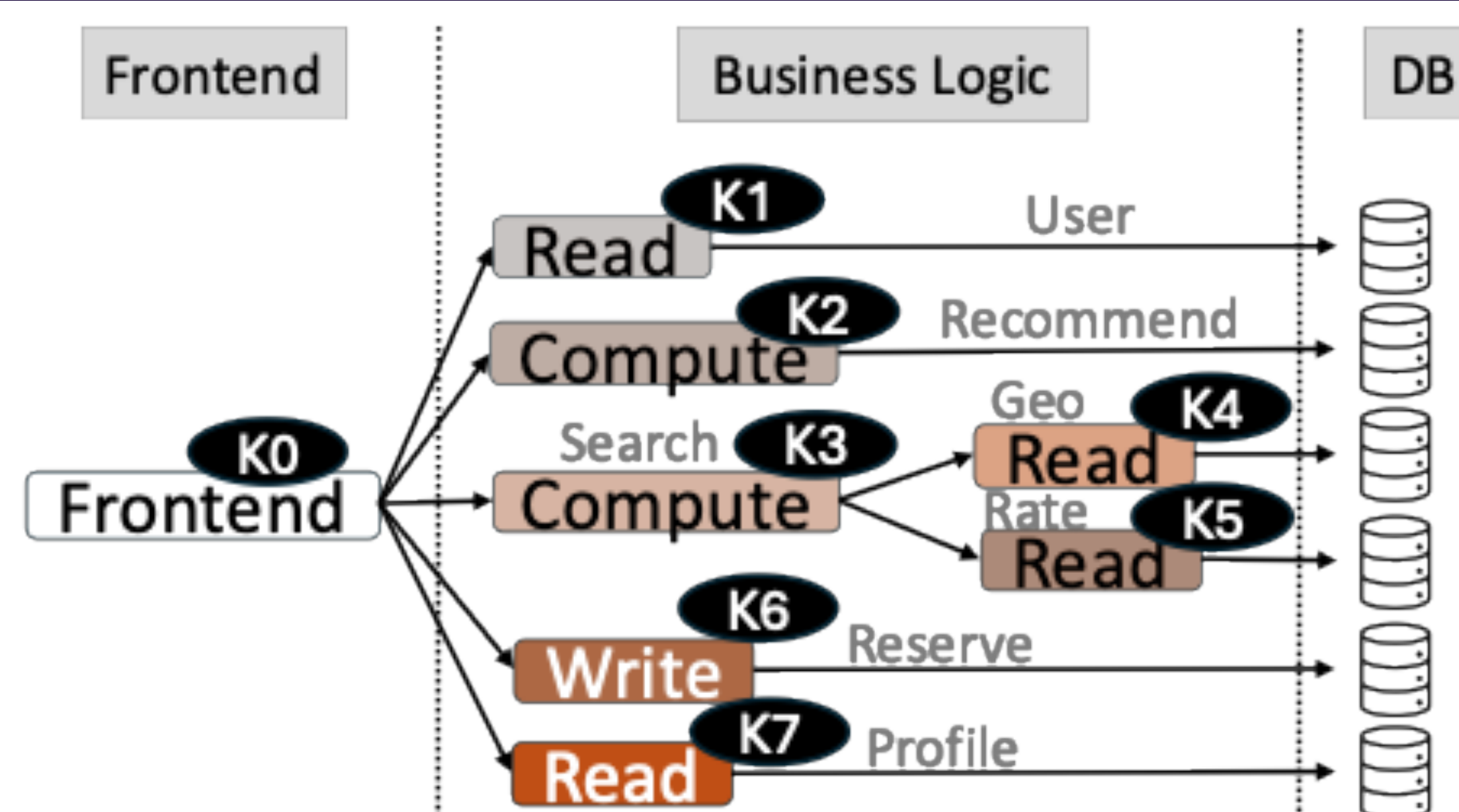


Fig. 2: MS Architecture of Hotel Reservation App.

Insight on MS Placement:

Which MS to move?

Early-activated MS subtrees

(amortize additional geographic latency over the remainder of the request execution)

Where to move them?

Strictly better regions (carbon & cost)
(converge faster & give better solutions)

C. System Design & Evaluation

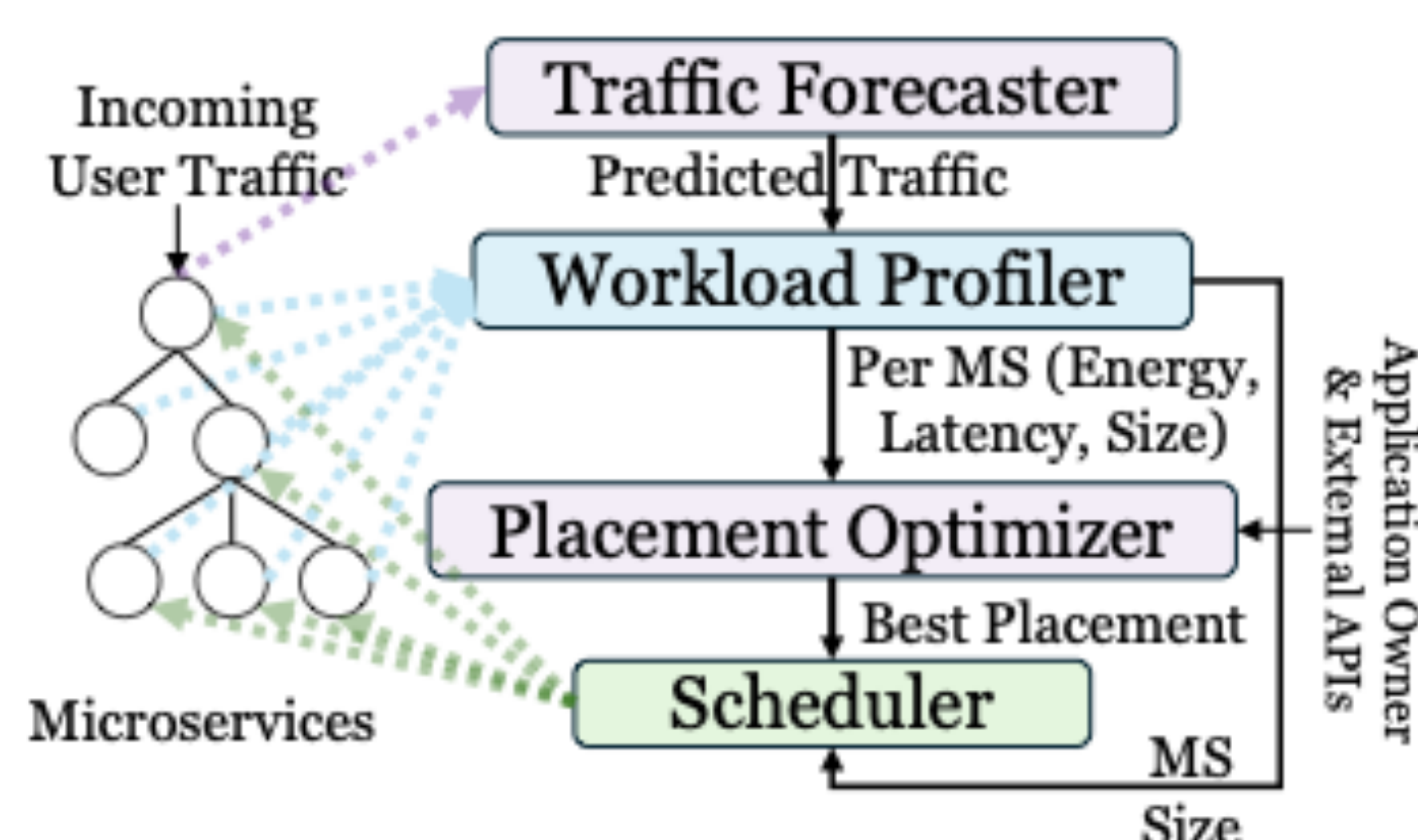


Fig. 3: System Components of Aceso.

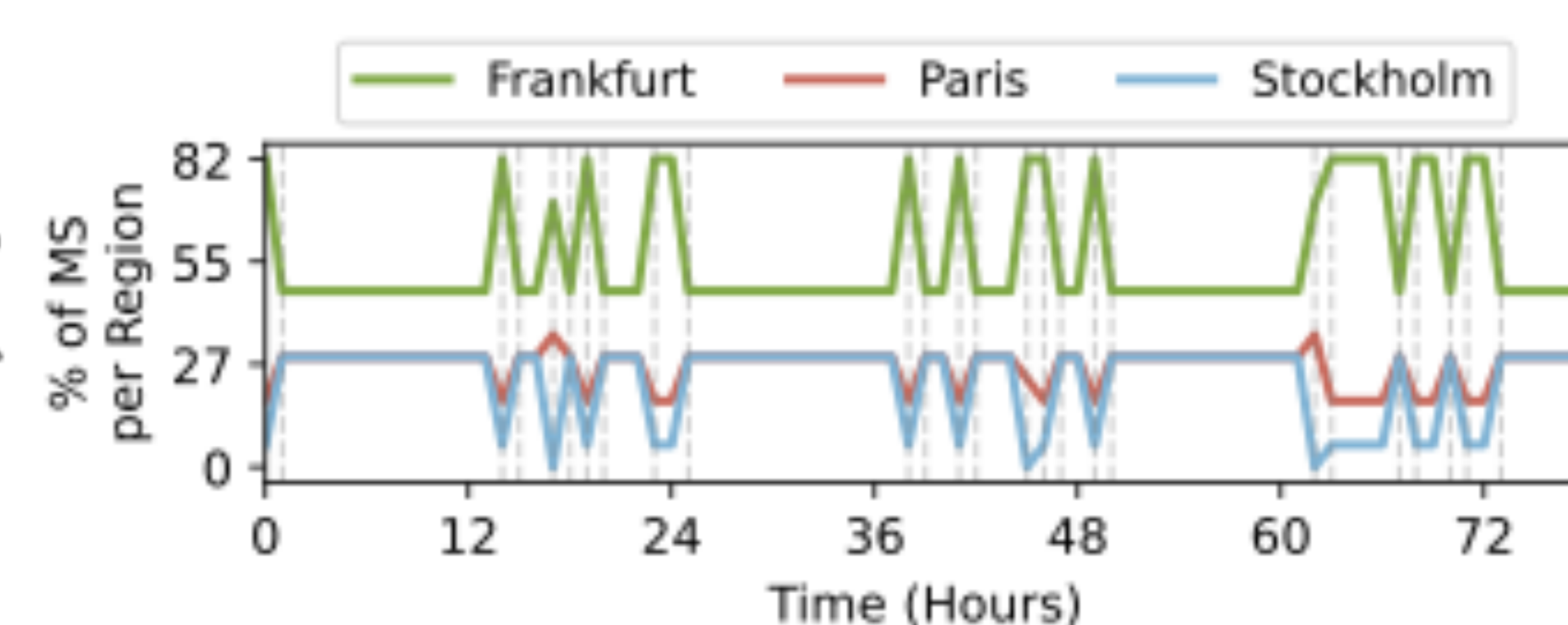


Fig. 4: MS Geo-distribution over time. Base region: Frankfurt.

Aceso solves an optimization problem to determine the map MS ↔ regions, effectively pruning the search space.

Aceso offloads 54% of MS to Paris and Stockholm and places them back to Frankfurt when traffic spikes to meet the latency goals.

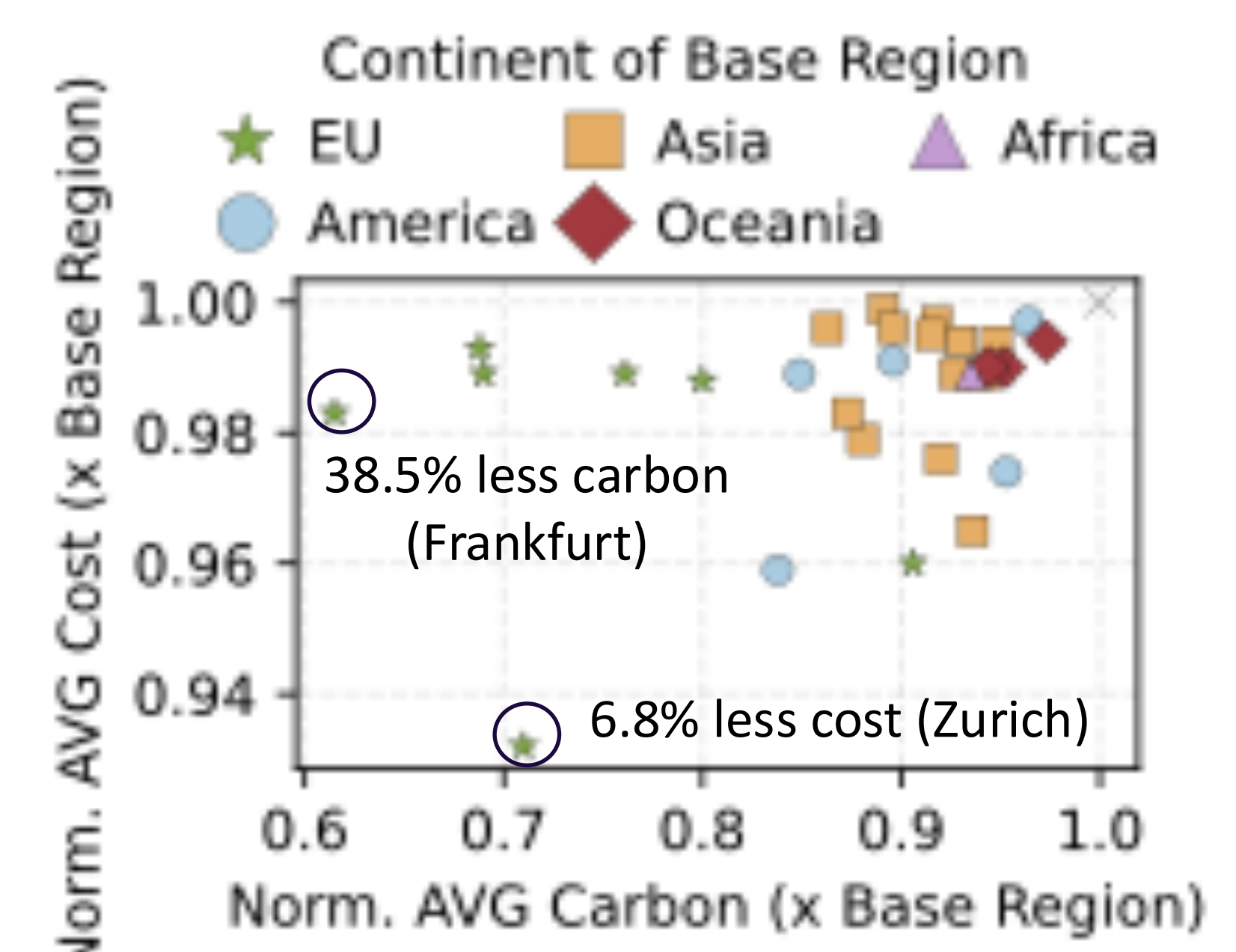


Fig. 5: Geographic Generalizability of Aceso.

EU-based SMEs achieve the largest carbon and cost reductions.

D. Comparison with Related Works

Related Work	Method
Caribou	Spatial Shifting for Serverless
Nautilus	MS Load Balancing (Single Region)
Genetic Alg. (GA)	Aceso w/o search space pruning
Linear Progr. (LP)	Exact optimization solver but very slow

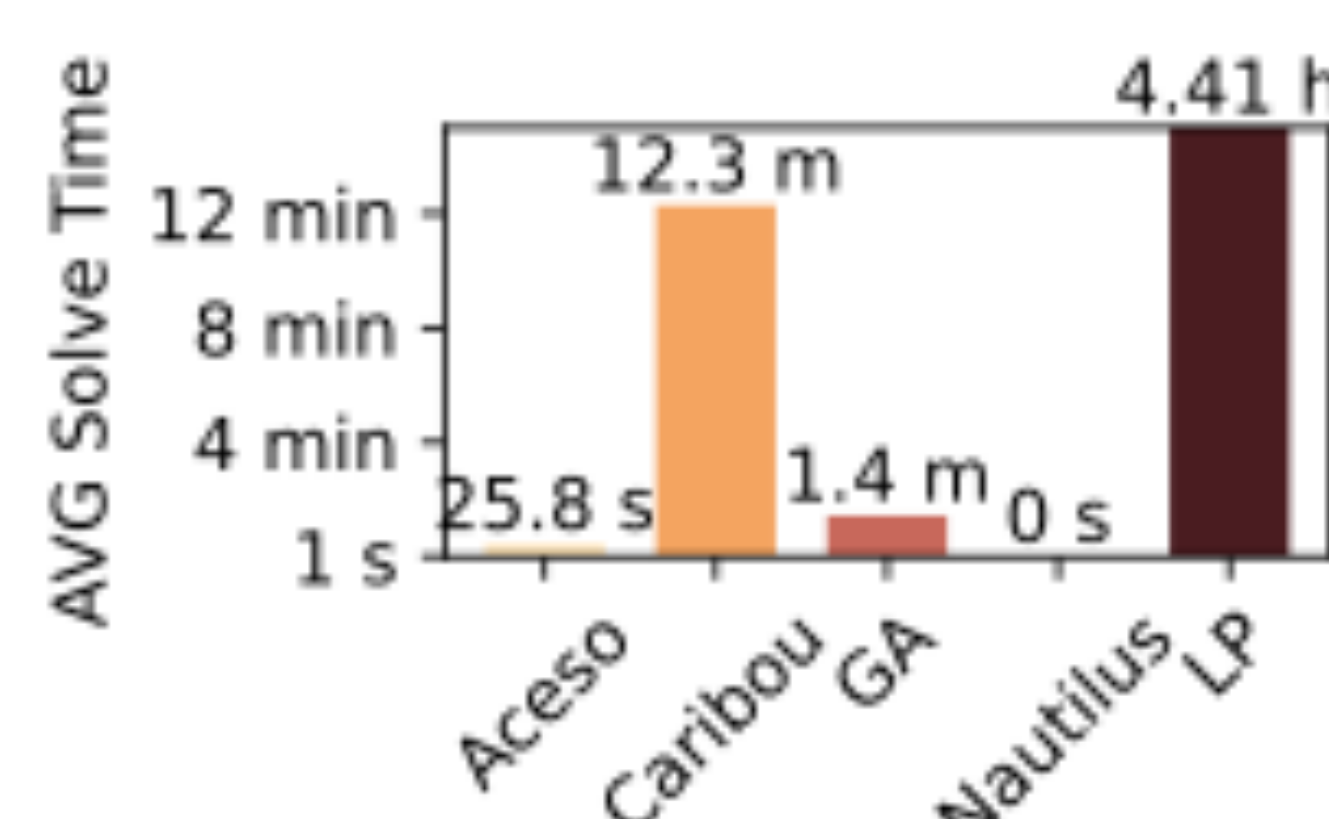


Fig. 6: Optimization solve time.

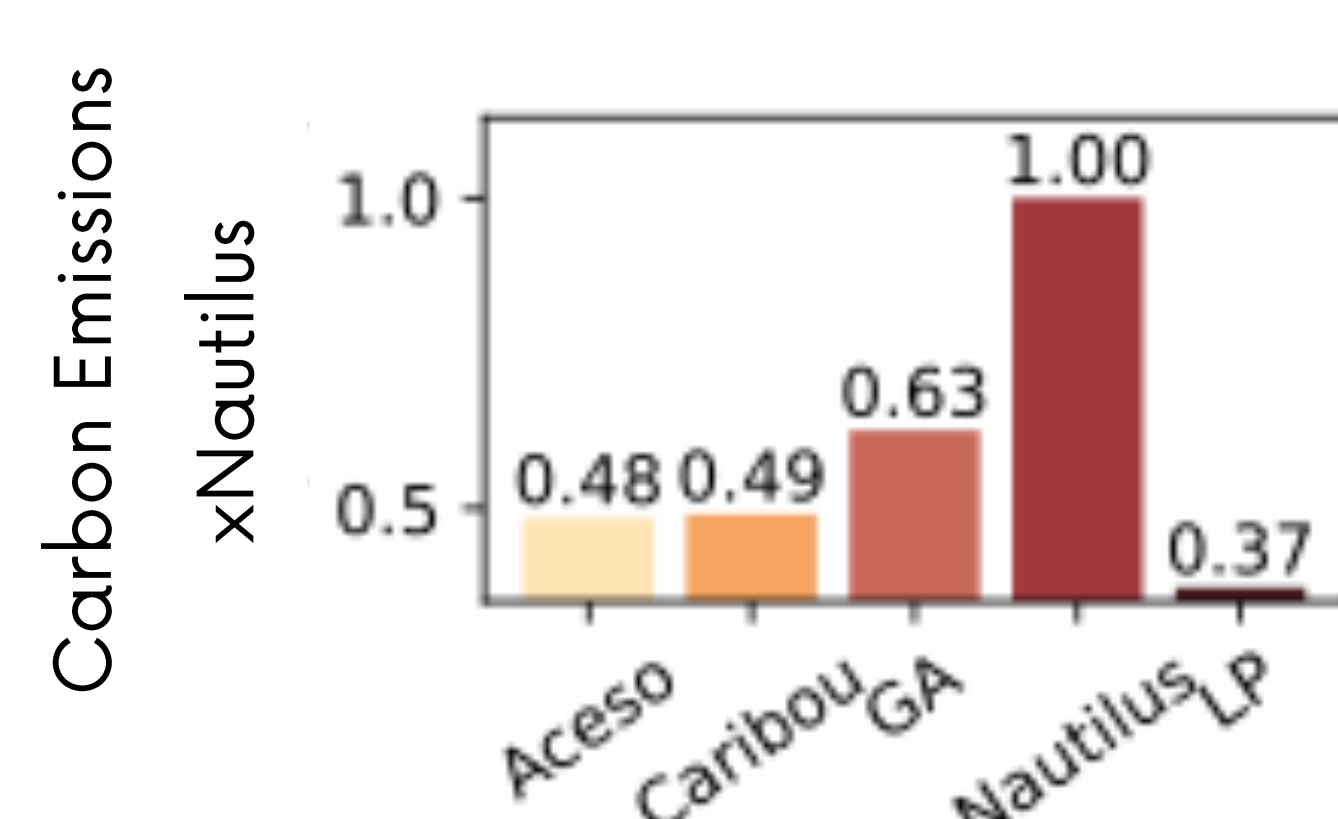


Fig. 7: Carbon Reduction (norm. Nautilus).

Takeaway: Aceso matches LP-level carbon savings while converging in seconds, far faster than other approaches, due to the use of GA and its insight-based search space pruning.

Full Paper to appear at **Middleware'26**

Aceso: Latency-Aware Geo-Distribution of Microservices for Cost-Effective Carbon Reduction

Aceso dynamically places MS across geo-constrained settings to reduce carbon emissions and cost while meeting latency constraints.

To be open sourced...